
Key Projects Overview

Feb. 2026

Overview List **AUTO**

Application	Product	Customer (OEM / End-user)	Seg.	Pot. [t/a]	SOP	Resp. (Sales/AD)
AAF (Active Air Flap) Drive Shaft	Grinova T XE 11603 (GF50) Black 9915	Kyungchang Industry (HKMC)	222	166	Q3 2027	PAIS / SEOJ
eWGA Gear	Grivory HTV-5H1 & HT2V-3H LF	UAES (BYD&GWM)	270	96	Q8.2027	WAHU / HEBA
Roof Rack Base	Grivory GV XE 16101 Black 9839 (GVL-5H HP)	Kemflo (RIVIAN)	230	21	Q2 2026	ZHER / FUVI
Vane Seal	Grinova H XE 11629 Black 9205 (H-GF30H NZ)	BorgWarner (VW/Hyundai)	270	50	05.2026	GUOH / XUJR
2K Cooling Coupler	Grilamid L 25A H X Black 9992	Blove Light (Li Auto)	222	65	08.2026	LTIN / HUAB

AAF (Active Air Flap) Drive Shaft / SEOJ

Material:

Grinova T XE 11603 (GF50) Black 9915

Requirements

- High stiffness at High temp.
(140°C x 1,000 hrs)
- High creep resistance
- High toughness

Customer: Kyungchang Industry

OEM/ End User: HKMC



Model: KA5

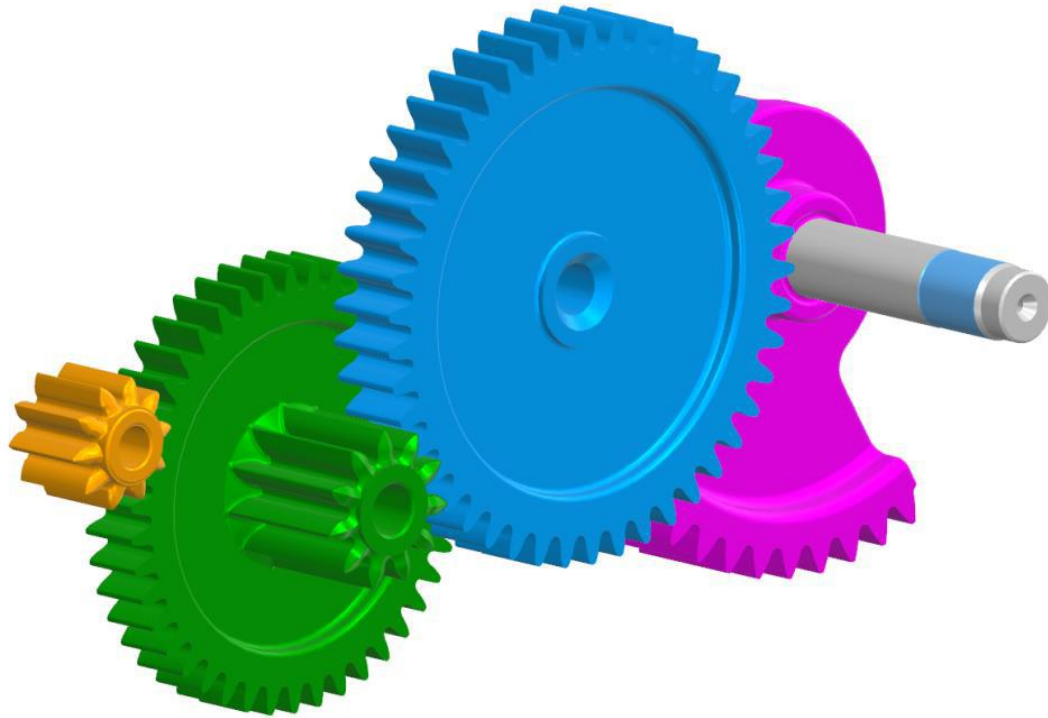
(Carnival 5th Generation)

AAF (Active Air Flap) Drive Shaft / SEOJ

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
AAF (Active Air Flap) Drive Shaft	Grinova T XE 11603 (GF50) Black 9915	Kyungchang Industry (HKMC)	222-260205-1	166	Q3 2027	PAIS / SEOJ
Advantages (<i>Technical Competition</i>)			Risks / Competitor Grade			
+ High stiffness at High temp. ($140^{\circ}\text{C} \times 1,000 \text{ hrs}$) + High creep resistance + High toughness			- Warping phenomenon <i>Plan to try for optimization through deviation of tool temp. upper & lower side</i>			
Actual Situation			Next Steps (<i>by When</i>)			Resp.
■ Proposed Grinova T products with consideration of requirements of application ■ Customer will check mechanical properties on application after molding trial.			■ Trial support (~30.06.26')			SEOJ

AD-TCS Key Project

eWGA Gear / HEBA



Material:

Grivory HTV-5H1 Black 9205

Grivory HT2V-3H LF Black 9205

Requirements

- Heat aging resistance at 160 °C
- Gear strength min. 650N
- Thermal shock @ -40°C ~ 160°C

Customer: UAES

OEM/ End User: BYD & GWM

AD-TCS Key Project

eWGA Gear / HEBA

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
eWGA Gear	Grivory HTV-5H1 & HT2V-3H LF	UAES (BYD&GWM)	270-251014-1	96	08.2027	WAHU / HEBA
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ Lower water absorption and density, compared with PA46			- Stanyl TW271F6 (PA46+PTFE+GF30) - Stanyl TW278F10 (PA46+PTFE+GF50) - Stanyl TW241F12 (PA46+GF60)			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
HT2V-3H LF was molded in the tooling of TW278F10 at molder. Dimension is over limit due to different GF content, but gear strength is better than PA46. (07.2025)			Add Grivory FE 10541 (HT1VS-5H PTFE) as alternative in PA46 replacement solution. (03.2026)			HEBA

AD-TCS Key Project

Roof Rack Base / ZHER

Material:
Grivory GV XE 16101 Black 9839
(GVL-5H HP)

Requirements

- High stiffness and strength at elevated temperature
- 500 hrs weathering resistance
- Outstanding creep resistance

Customer: Kemflo

OEM/ End User: RIVIAN



AD-TCS Key Project

Roof Rack Base / ZHER

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Roof Rack Base	Grivory GV XE 16101 Black 9839 (GVL-5H HP)	Kemflo (RIVIAN)	230-250812-2	21	Q2 2026	ZHER / FUVI
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ High stiffness at elevated temperature + Outstanding creep resistance + Superior energy absorption			- No			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
■ FEA done, GVL-5H passed with a factor of safety and the color was approved. ■ Mold trial done, functional test of these parts is on-going, the test result will be finished by 28th Feb. 2026.			■ Follow up the test result and Trial support (28th.02.2026)			ZHER

AD-TCS Key Project

Vane Seal / XUJR

Material:

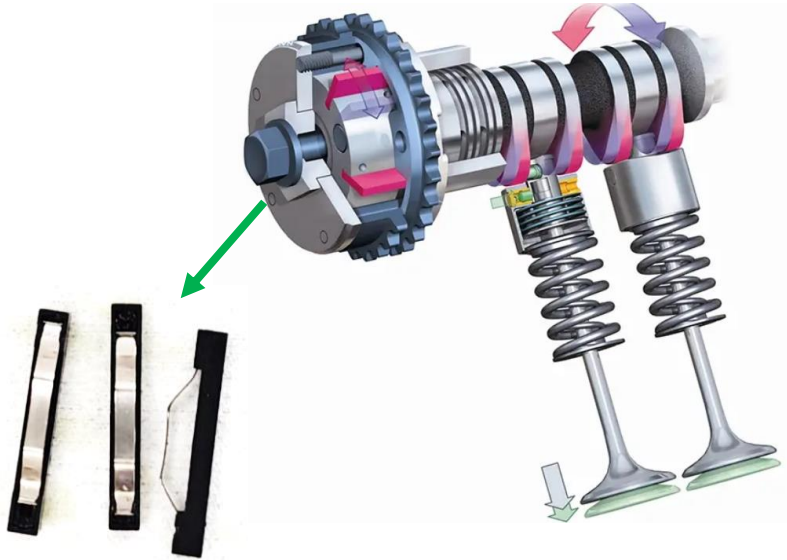
Grinova H XE 11629 Black 9205
(H-GF30H NZ)

Requirements

- Engine Oil resistance at 120°C
- Ultrasonic welding
- Excellent wear resistance

Customer: BorgWarner

OEM/ End User: VW & Hyundai



AD-TCS Key Project

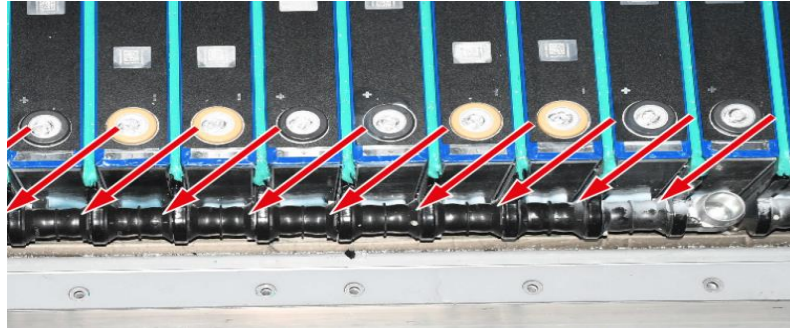
Vane Seal / XUJR

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Vane Seal	H XE 11629 Black 9205 (H-GF30H NZ)	BorgWarner (VW & Hyundai)	270-250710-2	50	05.2026	GUOH / XUJR
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ Success stories about PEEK replacement + Good ultrasonic welding property + Excellent oil and wear resistance			- PEEK-CF15 PTFE modified			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
■ Ultrasonic welding validation done ■ PV test and abrasion test of parts length 20 mm pass ■ New mold kick off			■ Support mold design for mass production and OTS parts delivery. (03.2026)			XUJR

AD-TCS Key Project

2K Cooling Coupler / HUAB

Material:
Grilamid L 25A H X Black 9992



Requirements

- **Balanced strength and toughness**
- **Coolant aging at 90°C for 600h**
- **Damp heat aging at 70°C/85% r.h., 85°C/85% r.h., 95°C/85%r.h. for 1,000h**

Customer: Blove Light

OEM/ End User: Li Auto

AD-TCS Key Project

2K Cooling Coupler / HUAB

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
2K Cooling Coupler	Grilamid L 25A H X Black 9992	Blove Light (Li Auto)	222-260119-1	65	08.2026	LTIN / HUAB
Advantages (<i>Technical Competition</i>)			Risks / Competitor Grade			
+ Success stories + Good hydrolysis resistance + Good toughness and strength			- No			
Actual Situation			Next Steps (<i>by When</i>)			Resp.
■ Material defined ■ Material test report submitted ■ Mold built ■ Material approval on going			■ Support mold trial on site. (03.2026)			HUAB

Overview List I&C

Application	Product	Customer (OEM / End-user)	Seg.	Pot. [t/a]	SOP	Resp. (Sales/AD)
E-scooter Display Housing	Grivory GV XE 11119 (GV-GFX50 V0)	Hualong (YADEA)	740	100	06.2026	MOSK ZHVI
Safety Valve of Electric Water Heater	Grivory GV-6H	Vanward (Haier)	410	75	05.2026	ZHLE LINR
HVAC Ball Valve	Grivory HT1V-5 FWA	JLT/Chuandibai (Grandsire)	410	50	02.2026	WEIC JIAJ
AGV Chassis	Grivory GV XE 16141 (GVCL-4H V0)	Chen'nuo (Geek+)	450	44	06.2026	GEJA ZHUP
Inner Bracket	Grilamid 2S XE 12211 (PA610-GF50)	Gengde (HUAWEI)	850	100	07.2026	JINE PEPE

E-scooter Display Housing / Hualong



Material:

Grivory GV XE 11119 (GV-GFX50 V0)

Requirements

- Low warpage than PC-GF, <0.5mm
- Thermal shock, -40 to 85°C, 75mins, 300cycles

Molder: Hualong

Tier: AISPEECH

OEM / End User: YADEA

E-scooter Display Housing / Hualong

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
E-scooter Display Housing	XE 11119 (GV-GFX50 V0)	Hualong (YADEA)	740-251212-1	100	06.2026	MOSK / ZHVI
Advantages (<i>Technical Competition</i>)			Risks / Competitor Grade			
+ Low warpage + Good mechanical strength			- Possible local competitors involved after production			
Actual Situation			Next Steps (<i>by When</i>)			Resp.
▪ AI replacement and problem-solving of PC/ABS due to waterproof failure and low heat resistance, 11.2025 ▪ 2t pilot run without problem, painting process optimization is ongoing to improve adhesion, 02.2026			▪ Support to resolve painting issue, 04.2026			ZHVI

Safety Valve of Electric Water Heater / Vanward



Reference picture

Material:
Grivory GV-6H BK

Requirements

- Brass replacement for >30% cost saving
- Reliable mechanical properties to withstand >60Nm torque force of G1/2' thread
- Static pressure 5.0 MPa, 1 min.

Customer: Vanward

OEM/ End User: Haier

Safety Valve of Electric Water Heater / Vanward

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Safety Valve	GV-6H	Vanward (Haier)	411-231107-1	75	05.2026	ZHLE LINR
Advantages (Technical Competition)			Risks / Competitor Grade			
+ Reliable mechanical properties + CATS & AD convince development			- Downgrade to PA66-GF50 with metal thread inserts			
Actual Situation			Next Steps (by When)			Resp.
- Reliable test of 3rd trial samples is still ongoing at Haier, 500kg was ordered to produce for oversea customer, 12.2025 - Finished all internal tests and ready to sell to the market (other OEMs), 02.2026			Obtain Haier test result, 30th Mar.			R. Lin

AD-TCS Key Project

Ball Valve / Grandsire



Material:
Grivory HT1V-5 FWA

Requirements

- High stiffness & strength
(burst pressure > 60bar @23°C)
- Food & water contact approved

Customer: JLT/ Chuandibai

Tier: Sansheng

End User: Grandsire

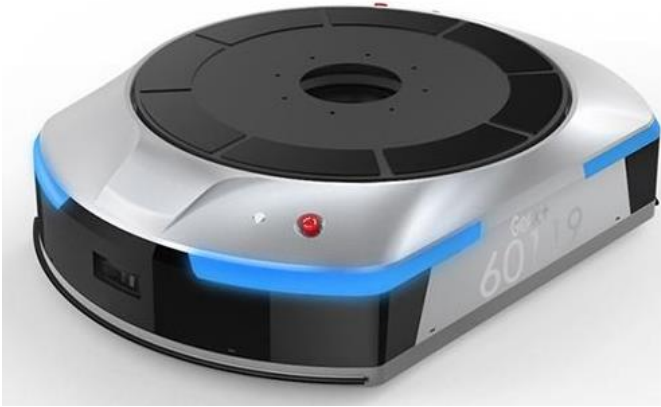
AD-TCS Key Project

Ball Valve / Grandsire

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Ball valve	Grivory HT1V-5 FWA	JLT/Chuandibai (Grandsire)	410-241011-1	50	02.2026	WEIC JIAJ, XUER
Advantages (Technical Competition)			Risks / Competitor Grade			
+ Technical support + High strength & stiffness			- No			
Actual Situation			Next Steps (by When)			Resp.
■ Chuandibai (1 st molder) part passed test and mass production started (9t material delivered), 11.2025 ■ Both JLT & Chuandibai in pilot production, around 20 ton material delivered, 02.2026			■ Track mass production status, 03.2026			JIAJ

AD-TCS Key Project

AGV Chassis / Geek+



Material:

Grivory GV XE 16141 (GVCL-4H V0)

Requirements

- 600,000 cycles lifting test @ 40kg
- 150,000 times 50mm height obstacle tests
- 100 times collisions @ 4m/s
- Chassis deflection < 3mm @ 40kg

Customer: Chennuo

End User: Geek+

AD-TCS Key Project

AGV Chassis / Geek+

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
AGV Chassis	XE 16141 (GVCL-4H V0)	Chennuo (Geek+)	450-250925-1	45	06.2026	GEJA ZHUP
Advantages (Technical Competition)			Risks / Competitor Grade			
+ Material performance (UL V0 & ESD) + Technical support (FEA, MFA, on-site trial)			- Insufficient cost saving			
Actual Situation			Next Steps (by When)			Resp.
■ New design AGV chassis, plastic solution has ~15% cost saving vs metal die-casting, 10.2025 ■ FEA result showed acceptable deformation (1.5 vs. spec 2.0mm) & stress level (Max. 59 MPa), 10.2025 ■ Tooling were ready and 8 pcs samples collected for function test, 02.2026			■ On-site check test status, 03.2026			ZHUP

Smart Phone Inner Bracket / Gengde



Material:

Grilamid 2S XE 12211 (PA610-GF50)

Requirements

- Good stiffness
- Lower warpage & dimensional stability
- Low Permittivity Dk

Customer: Gengde

OEM/ End User: HUAWEI

Smart Phone Inner Bracket / Gengde

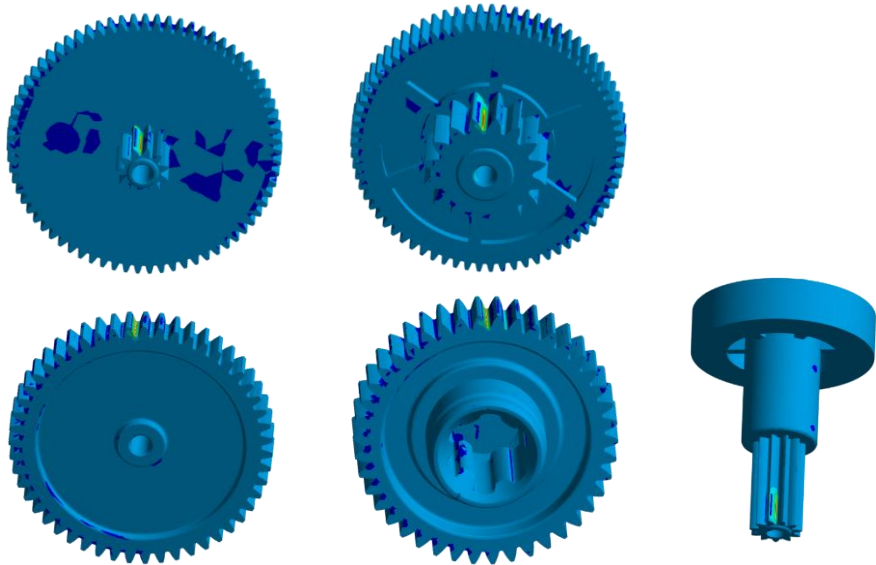
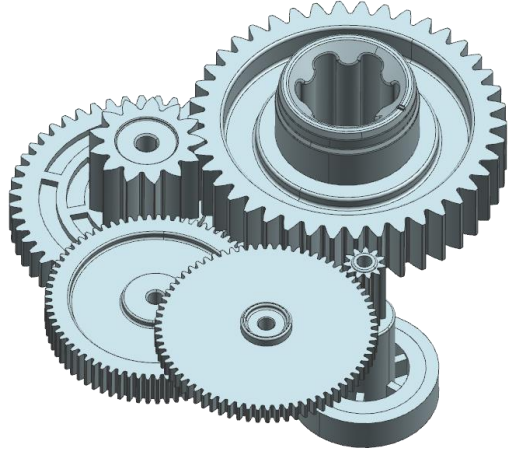
Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Inner Bracket	Grilamid 2S XE 12211 (PA610- GF50)	Gengde (HUAWEI)	850-260203-1	100	07.2026	JINE PEPE
Advantages (Technical Competition)			Risks / Competitor Grade			
+ Low Permittivity Dk + Good stiffness & dimensional stability			- Syensqo Kalix 2545 - Commercial compete with Syensqo Kalix 2545			
Actual Situation			Next Steps (by When)			Resp.
▪ Project initiated for the new generation of smart phone - Pura 90. Sample done by customer, test is ongoing, 02. 2026			▪ Obtain test results, 04. 2026			PEPE

Overview List CATS

Application	Product	Customer (OEM / End-user)	Seg.	Pot. [t/a]	SOP	Resp. (Sales/AD)
Actuator Gear	Grivory HT2VZ-33H/HT2V-3H LF/HT1VS-5H1/HTV-4H1	Temb (Chery&Geely)	222	50	Dec. 2026	WANH/HULO, LIDI
Dyer Housing	Grivory HTV-5H1	Anzhi (Geely)	270	30	Jun. 2026	XUEV/LIPE, XUER
Oil Jet Pipe	Grivory HTV-4H1	Hongqi (Hongqi)	242	24	Jan. 2027	SONG/ZHZA, WABR
Coffee Machine BU Shell	Grivory GVX-5H	Providence (LAVAZZA)	480	85	Dec. 2026	YAOE/HELU, YANC
Front Cover	Grilamid LV-30H V0	Zhengda (Stryker)	710	30	May 2025	CHNC/LICR, JIJO

AD-TCS Key Project

Actuator Gears / Temb



Material:

Grivory HT2VZ-33H / Grivory HT2V-3H LF
Grivory HT1VS-5H1 / Grivory HTV-4H1

Requirements

- High strength and fatigue performance under various temperatures (*test loads of 2Nm@-40 °C, 1.5Nm@23 °C, 1Nm@120 °C*)
- High temperature resistance

Customer: Temb

OEM/ End User: Chery & Geely

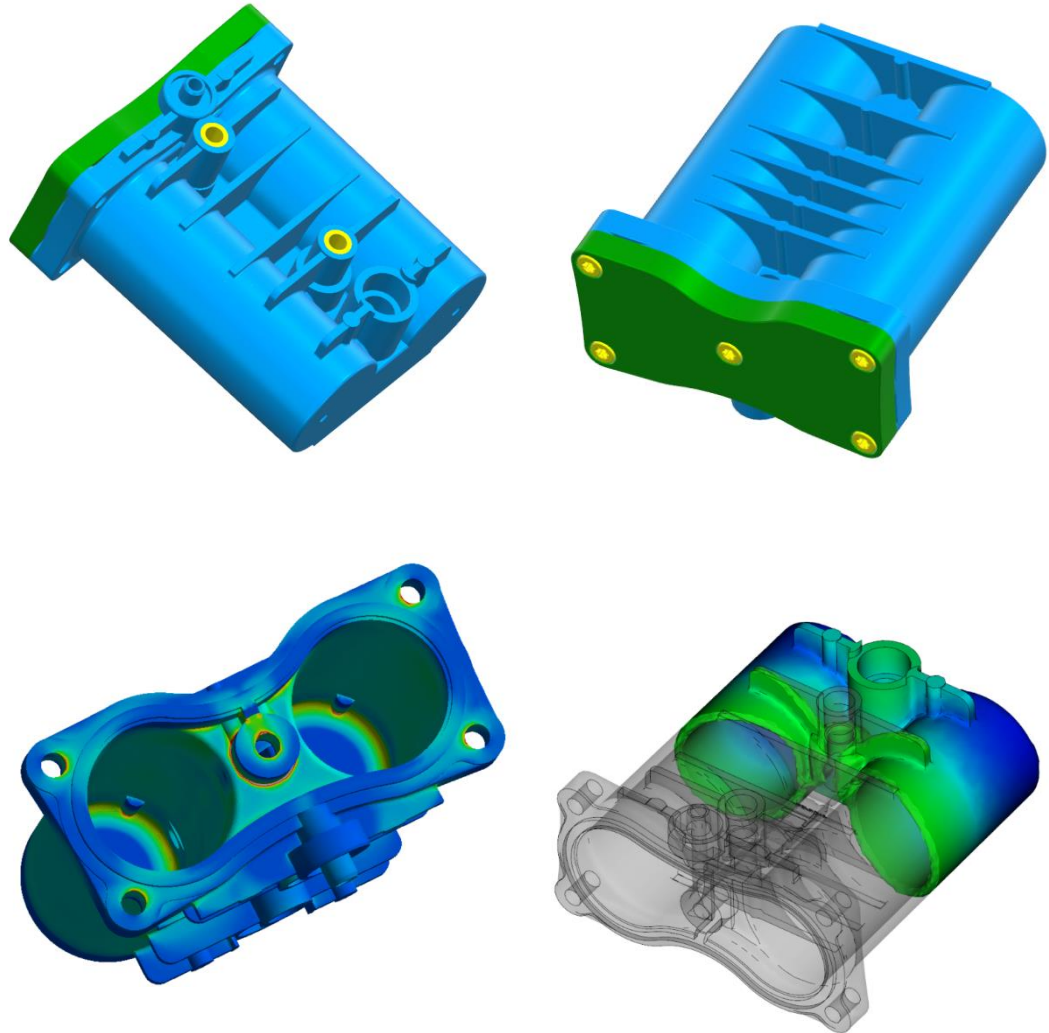
AD-TCS Key Project

Actuator Gears / Temb

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Actuator Gear	Grivory HT2VZ-33H Grivory HT2V-3H LF Grivory HT1VS-5H1 Grivory HTV-4H1	Temb (Chery&Geely)	222-260114-1	50	Dec. 2026	WANH/HULO, LIDI
Advantages (Technical Competition)			Risks / Competitor Grade			
+ High strength at high temperature + Cost saving compared to PA46 + FEA support			- Envalior Stanyl TW241F10 (PA46 GF50)			
Actual Situation			Next Steps (by When)			Resp.
■ FEA of 5x gears done. ■ Report with design suggestions provided to customer.			■ Follow up the customer's decision by Mar. 2026			LIDI

AD-TCS Key Project

Dryer Housing / Anzhi



Material:
Grivory HTV-5H1

Requirements

- High strength & stiffness at high temperature (*burst pressure of 36bar@23 °C and 20bar@150 °C*)
- High temperature resistance

Customer: Anzhi

OEM/ End User: Geely

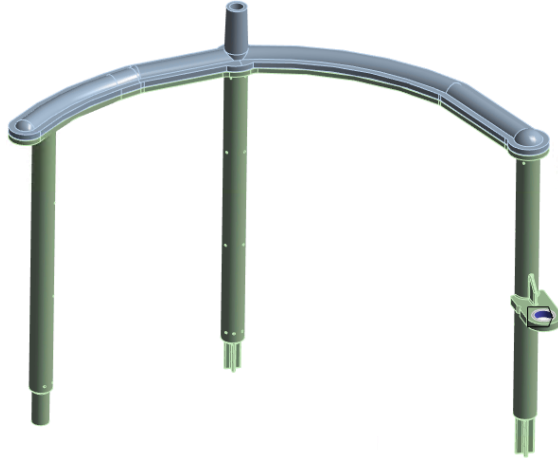
AD-TCS Key Project

Dryer Housing / Anzhi

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Dryer Housing	Grivory HTV-5H1	Anzhi (Geely)	270-251204-1	30	Jun. 2026	XUEV/LIPE, XUER
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ High strength & stiffness at high temperature + FEA support			- Insufficient weld line strength			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
- FEA performed to find high stress areas. - Weld lines distribution investigated with MFA. - CAE report with design suggestions provided.			Follow up tooling by Apr. 2026			XUER

AD-TCS Key Project

Oil Jet Pipe / Hongqi



Material:
Grivory HTV-4H1

Requirements

- **High strength & stiffness at High temperature** (*burst pressure >4bar @120 °C*)
- **High temperature resistance** (*oil aging at 150 °C, 1'008 hours*)

Customer: Hongqi

OEM/ End User: Hongqi

AD-TCS Key Project

Oil Jet Pipe / Hongqi

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Oil Jet Pipe	Grivory HTV-4H1	Hongqi (Hongqi)	242-260123-1	24	Jan. 2027	SONG/ZHZA, WABR
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ Cost and weight saving compared to metal + Success story with Grivory HT + FEA support			- No			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
■ Feasibility study with FEA, done. ■ FEA report with design suggestions provided to customer.			■ Follow up customer's decision and support MFA if needed by Mar. 2026			WABR

Coffee Machine BU Shell / Providence

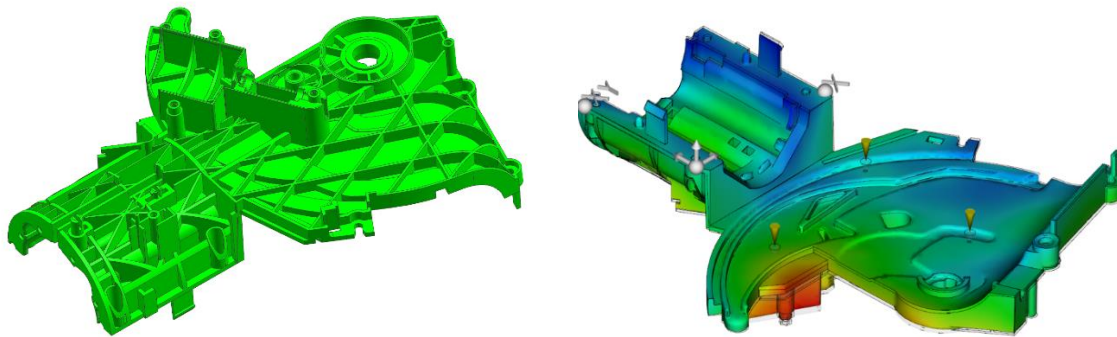


Reference

Material:
Grivory GVX-5H

Requirements

- **High warpage performance**
(out of flatness ≤ 0.2 mm)
- **High strength**
(burst pressure of 35 bar@23 °C)



Customer: Providence

OEM/ End User: LAVAZZA

Coffee Machine BU Shell / Providence

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Coffee Machine BU Shell	Grivory GVX-5H	Providence (LAVAZZA)	480-260113-2	85	Dec. 2026	YAOE/HELU, YANC
Advantages (<i>Technical Competition</i>)			Risks / Competitor Grade			
+ High strength & stiffness + Very low warpage + MFA support			- PA66 GF30			
Actual Situation			Next Steps (<i>by When</i>)			Resp.
■ Various gating scenarios investigated with MFA to improve warpage. ■ Gating solutions explained to customer. ■ Soft tool is under preparation.			■ Follow up tooling by Apr. 2026			YANC

AD-TCS Key Project

Front Cover / Zhengda

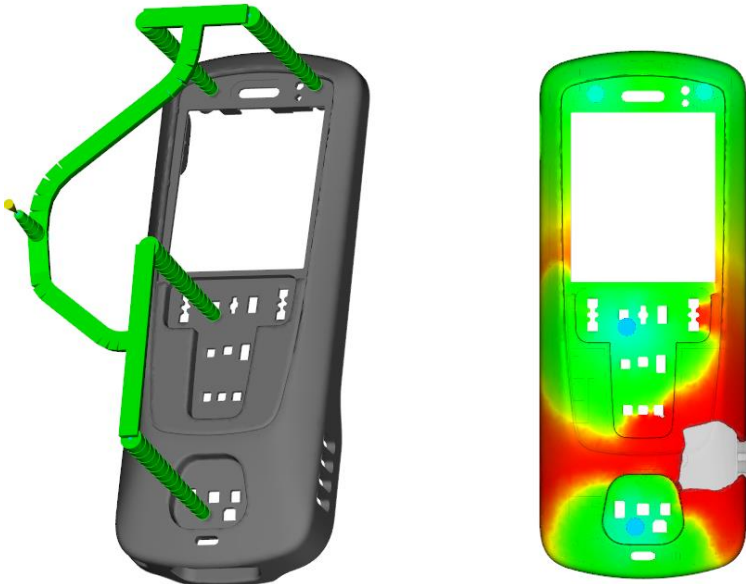
Material:
Grilamid LV-30H V0

Requirements

- Chemical resistance
- High flow ability
- Low warpage

Customer: Zhengda

OEM/ End User: Stryker



AD-TCS Key Project

Front Cover / Zhengda

Application	Material	Customer (OEM, End-user)	Project No. or Segment	Pot. [t/a]	SOP	Resp. [Sales/AD]
Front Cover	Grilamid LV-30H V0	Zhengda (Stryker)	710-240218-1	30	May 2025	CHNC/LICR, JJO
Advantages <i>(Technical Competition)</i>			Risks / Competitor Grade			
+ High dimensional stability + Good chemical resistance and UV stability + Technical support			- Air trap due to uneven wall-thickness			
Actual Situation			Next Steps <i>(by When)</i>			Resp.
■ Air trap with surface issue detected at T0. ■ Gating optimized with MFA. ■ MFA report provided.			■ Follow up tool modification by Mar. 2026			JJO